

ABDUL WASAY WAQAS

Full-Stack AI Engineer

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PROFESSIONAL SUMMARY

Full-Stack AI Engineer building end-to-end AI products — real-time voice/LLM pipelines, FastAPI/Django backends, and React/Next.js frontends. Shipped a production voice AI platform serving 40+ concurrent sessions and 1,000+ real-time users, cutting latency 55% (2200ms to under 1000ms). Own features across the full stack — data model, API, AI pipeline, tests, and UI — deployed on AWS via CI/CD.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Rust, SQL, Bash, HTML, CSS

AI & ML: LLM Orchestration, RAG, Prompt Engineering, Tool & Function Calling, Vector Search, Embeddings, STT/TTS/VAD, Speaker Diarization, LiveKit Agents, Multi-Provider (OpenAI, Anthropic, Groq, Deepgram, ElevenLabs, Cartesia, Google, Azure)

Frontend: React.js, Next.js, TypeScript, Tauri (Rust desktop), Redux Toolkit, TanStack Query, Tailwind CSS, Radix/shadcn, Recharts, Responsive UI, SEO

Backend: FastAPI, Django, Django REST, Async Python, REST APIs, WebSockets, WebRTC, Socket.IO, SQLAlchemy, Pydantic, Celery/arq, Microservices, JWT & OAuth

Databases & Cloud: PostgreSQL, MongoDB, MySQL, Redis, Qdrant (Vector DB), Firebase; AWS (EC2, S3), Docker, Nginx, FFmpeg, GitHub Actions, CI/CD

Testing: Pytest, Playwright (E2E), Unit & Integration Testing

WORK EXPERIENCE

Full-Stack AI Engineer, Convoi AI • go.convoi.ai

Jan 2025 – Present

- Architected and shipped a production voice AI platform on LiveKit (40+ concurrent sessions) with STT–LLM–TTS pipelines integrating 15+ providers (OpenAI, Anthropic, Deepgram, ElevenLabs, Cartesia, Groq, Google, Azure, AssemblyAI).
- Cut end-to-end voice agent latency 55% (2200ms to under 1000ms) via real-time STT-to-TTS orchestration, streaming token generation, and VAD tuning.
- Built a FastAPI backend (PostgreSQL, Redis/arq, Qdrant vector DB, Socket.IO) exposing 100+ REST endpoints, with runtime tool/function calling and RAG over Qdrant for multi-turn conversations.
- Owned the frontend: a Next.js 16 admin CMS with 150+ responsive components (analytics, billing, RBAC, multi-tenant management) wired to the backend APIs.
- Designed an async subscriber microservice capturing 50+ concurrent LiveKit recordings via FFmpeg audio merging (WAV to Opus OGG) with reliable S3 upload and safe finalization.
- Implemented JWT/OAuth + RBAC scaling to 1,000+ concurrent real-time users; integrated SIP telephony and WhatsApp Business workflows.
- Wrote Pytest + Playwright tests and deployed to AWS EC2 via GitHub Actions CI/CD; hardened AI reliability with retries, circuit breakers, and fallback model routing.

Freelance Full-Stack & Web Developer, Fiverr (Remote)

2022 – 2023

- Delivered full-stack web apps and SEO-optimized sites for international clients, focused on performance, usability, and on-time delivery.

JavaScript Intern, Qwerty Experts

Jul 2023 – Sep 2023

- Built responsive React/JavaScript UI components and shipped production frontend features under code review and Git-based workflows.

PROJECTS

HintMint — Cross-Platform Desktop AI Assistant • Tauri (Rust), React, TypeScript, LLM, RAG • hintmint.io

- Cross-platform desktop AI assistant (Tauri/Rust + React) shipped at production scale and published on the Microsoft Store for Windows.
- Built an LLM-powered response pipeline with prompt engineering and RAG for contextual accuracy of real-time AI outputs.
- Owned the product end-to-end across the native Rust layer, the React UI, and the AI backend.

Diarization Studio — Speaker Separation & Call Analysis • Next.js, Django REST, Deepgram, Groq LLM, PyAV

- Full-stack app that diarizes calls into per-speaker transcripts + isolated audio, with talk-time analytics, SRT/VTX/JSON export, an LLM call-analysis feature (Groq), and translation.

Convoi CMS & Marketing Website • Next.js, React, TypeScript, Tailwind CSS • convoi.ai

- Built 150+ reusable, responsive React components across an SEO-optimized marketing site and a data-dense multi-tenant admin console sharing one design system.

Real-Time Streaming & AI Voice Chat Platform • WebRTC, LiveKit, React, WebSockets, STT, LLM

- Built a real-time streaming app (WebRTC + LiveKit) with an integrated AI voice chatbot — speech-to-text feeding LLM-based responses — over a responsive React frontend with live chat.

EDUCATION

BS Computer Science — Government College University, Lahore

2021 – 2025

FSc Pre-Engineering — Government College, Lahore